

SML A4

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1 Question 1: AdaBoost with Decision Stumps

1.1 Methodology and Math

The MNIST dataset was filtered to retain only digits **4** and **9**, relabelled as $y \in \{-1, +1\}$ respectively. Images were flattened and normalised to $[0, 1]$. 1000 samples from each class were held out for validation. PCA was then fit *only* on the remaining training data and used to reduce the dimensionality to $p = 5$.

AdaBoost builds a strong classifier as a weighted vote of T weak learners (decision stumps):

$$H(x) = \text{sign}\left(\sum_{t=1}^T \alpha_t h_t(x)\right)$$

At each round t , a stump h_t is fit by minimising the **weighted misclassification error**:

$$\varepsilon_t = \sum_{i=1}^n w_i \cdot \mathbf{1}[h_t(x_i) \neq y_i]$$

The stump weight and the weight update rule are:

$$\alpha_t = \frac{1}{2} \ln\left(\frac{1 - \varepsilon_t}{\varepsilon_t}\right) \quad w_i \leftarrow w_i \cdot \exp(-\alpha_t y_i h_t(x_i)), \quad \text{then normalise}$$

Weights start uniform at $w_i = \frac{1}{n}$. Misclassified samples receive higher weight in the next round, forcing subsequent stumps to focus on hard examples.

Decision Stump: For each of the 5 PCA dimensions, candidate thresholds are midpoints between consecutive unique values (up to 1000 randomly sampled per dimension). Both polarities (+1 and -1) are tried:

$$h(x) = \begin{cases} +\text{polarity} & \text{if } x_f \leq \theta \\ -\text{polarity} & \text{otherwise} \end{cases}$$

The stump minimising ε_t is selected at each round.

1.2 Outputs and Results

- **Data Shape:** Train (after split): see code output. Val: 2000 samples. Test: filtered 4 vs 9.
- **Number of Stumps:** $T = 300$
- **Best Iteration:** The iteration achieving the highest validation accuracy.
- **Test Accuracy:** Evaluated at the best iteration on the held-out test set.

1.3 Plots and Explanations

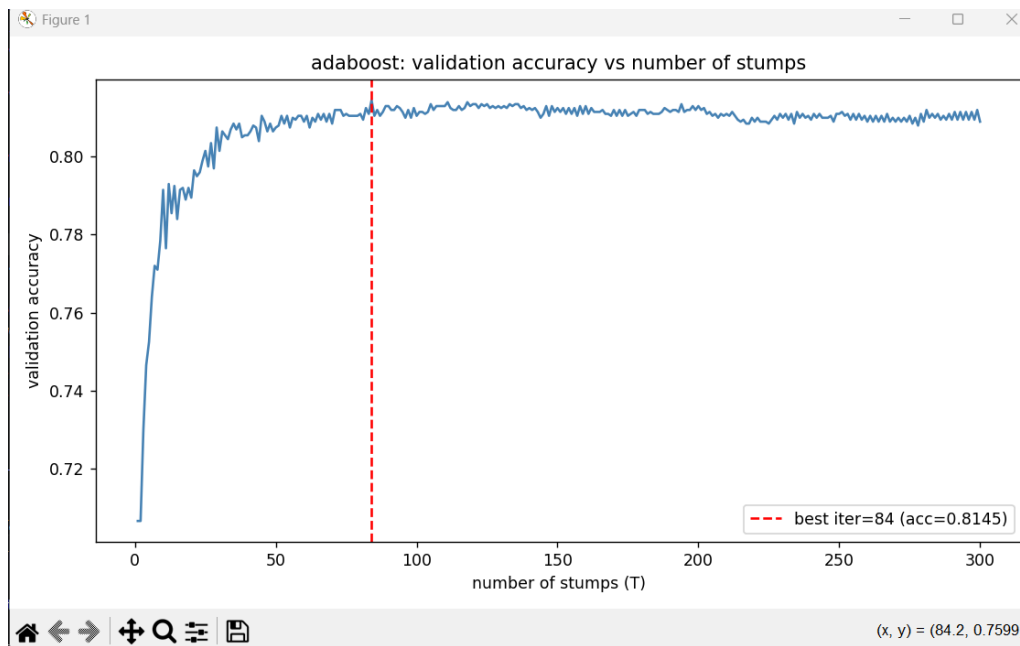


Figure 1: AdaBoost: Validation Accuracy vs. Number of Stumps

Observation: Validation accuracy rises sharply in the first few dozen rounds as the ensemble corrects the most common errors. It then plateaus and may slightly decline at very high T due to overfitting of the weighted distribution to training noise. The red dashed line marks the best iteration used for test evaluation.

Observation: A single stump achieves accuracy only slightly above chance for this binary task. The ensemble rapidly closes the gap toward near-perfect separation of digits 4 and 9, which are visually very different classes.

2 Question 2: Gradient Boosting with Decision Stumps

2.1 Methodology and Math

The same dataset and PCA pipeline from Question 1 are reused. Gradient Boosting builds an additive model:

$$F(x) = \sum_{t=1}^T \eta \cdot h_t(x)$$

where η is the learning rate and each h_t is a regression stump fit to the **negative gradient** of the loss at the current prediction.

Residuals (L2 loss): For squared loss $L = \frac{1}{2}(y - F(x))^2$, the negative gradient is simply the residual:

$$r_i = y_i - F(x_i)$$

Label update (absolute loss): As specified in the assignment, the stump at each round is fit to the **sign of the residuals**, which is the negative gradient of the absolute loss $L = |y - F(x)|$:

$$\tilde{y}_i = \text{sign}(r_i) \in \{-1, +1\}$$

Regression Stump (SSR): The stump minimises the Sum of Squared Residuals:

$$\text{SSR} = \sum_{i \in L} (r_i - \bar{r}_L)^2 + \sum_{i \in R} (r_i - \bar{r}_R)^2$$

where $\bar{r}_L$ and $\bar{r}_R$ are the mean residuals in the left and right leaves. The leaf prediction is the mean of the (absolute-loss) labels in that region.

Update rule:

$$F(x) \leftarrow F(x) + \eta \cdot h_t(x)$$

Evaluation: After each round, MSE is computed on train, validation, and test sets:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - F(x_i))^2$$

2.2 Outputs and Results

- **Main run:** $\eta = 0.01$, $T = 300$ stumps.
- **Best Iteration:** Iteration achieving lowest validation MSE.
- **Learning rates compared:** $\eta \in \{0.001, 0.01, 0.1, 0.2, 0.5, 1.0\}$

2.3 Plots and Explanations

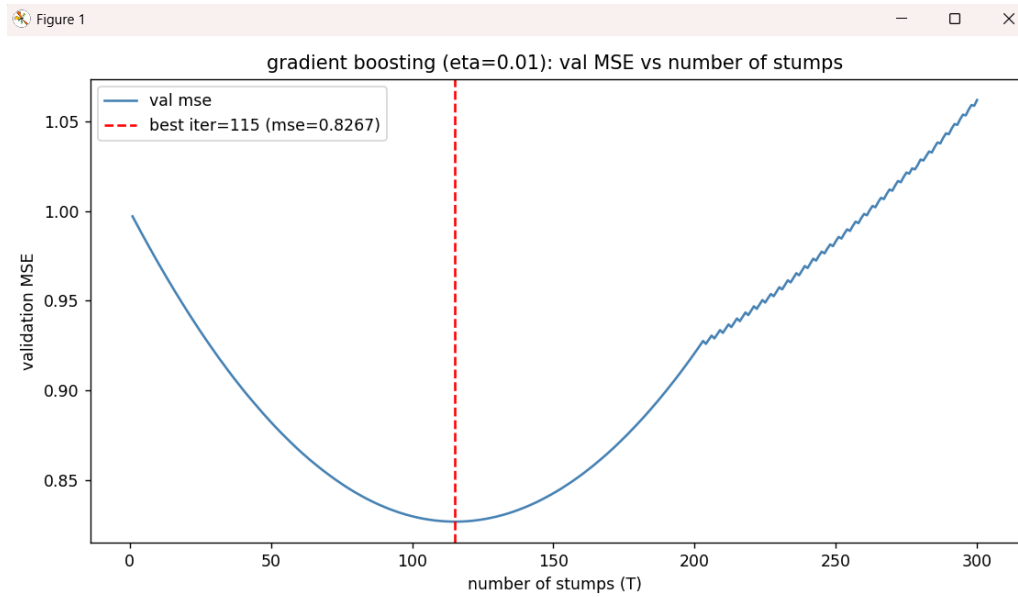


Figure 2: Gradient Boosting ($\eta = 0.01$): Validation MSE vs. Number of Stumps

Observation: With $\eta = 0.01$, the validation MSE decreases gradually and smoothly, reaching its minimum before slowly rising again as the model begins to overfit the training residuals.

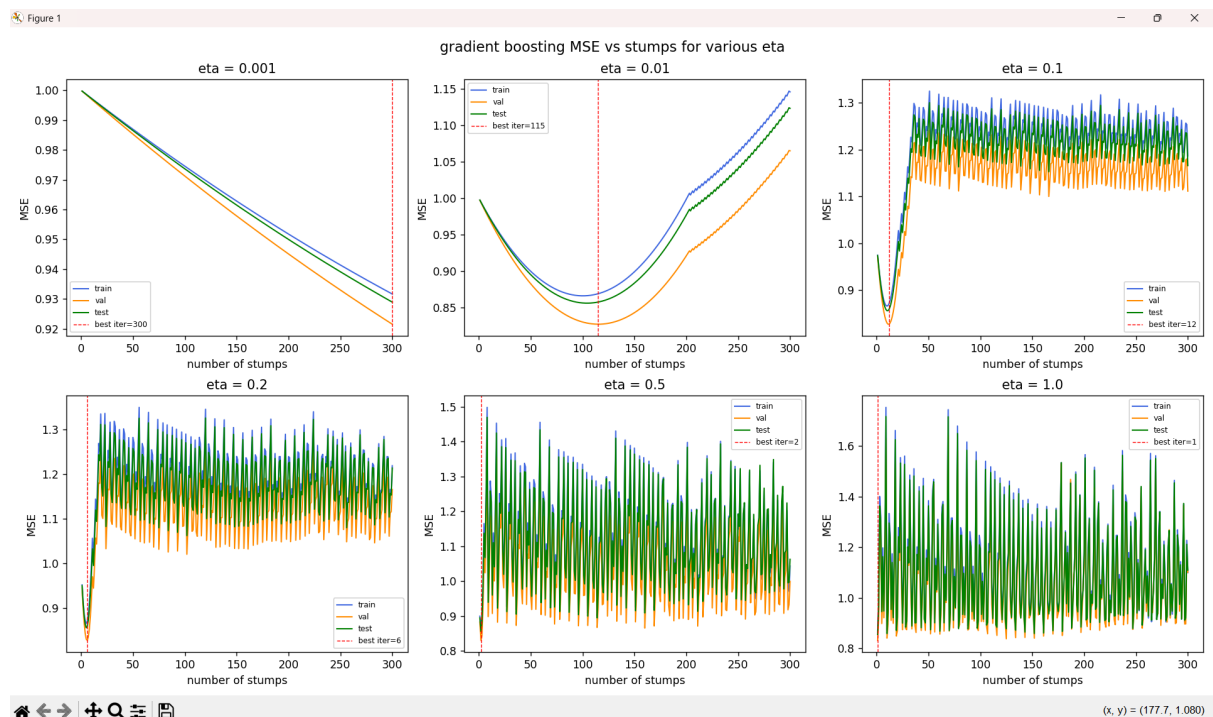


Figure 3: MSE vs. Number of Stumps for Various Learning Rates η

Observation (Learning Rate Trends):

- **Very small** $\eta = 0.001$: Convergence is extremely slow. Even after 300 stumps the MSE remains high because each stump contributes a negligible update. The model underfits within the budget of $T = 300$.
- **Small** $\eta = 0.01$: Smooth, gradual descent. Train, val, and test MSE track each other closely, indicating good generalisation. This is the most balanced choice.
- **Moderate** $\eta \in \{0.1, 0.2\}$: Convergence is faster; the best iteration arrives earlier. However, train MSE drops more steeply than val MSE after the peak, indicating slight overfitting beyond the optimal iteration.
- **Large** $\eta \in \{0.5, 1.0\}$: The model converges in very few iterations but overshoots the optimum, causing val and test MSE to rise sharply after just a handful of stumps. Generalisation is the worst among all rates.

Key Insight: Smaller η requires more stumps but gives a smoother loss landscape and better generalisation. Larger η risks divergence and early overfitting. This is the standard bias-variance trade-off along the boosting axis: more rounds at lower step size is preferable.

3 Question 3: Rosenblatt Perceptron

3.1 Methodology and Math

Two synthetic 2D datasets were generated:

- **Dataset A:** Class -1 : $X \sim \mathcal{N}\left(\begin{bmatrix} -3 \\ -3 \end{bmatrix}, I\right)$, Class $+1$: $X \sim \mathcal{N}\left(\begin{bmatrix} 3 \\ 3 \end{bmatrix}, I\right)$. 200 samples per class.
- **Dataset B:** Same means, but covariance $= 3I$. 200 samples per class.

Both datasets were split 70–30 into train and test sets. All weights and biases were initialised to zero.

Rosenblatt Perceptron is a linear classifier:

$$f(x) = \text{sign}(\mathbf{w}^\top x + b)$$

Update rule (applied only on misclassified samples where $y_i(\mathbf{w}^\top x_i + b) \leq 0$):

$$\mathbf{w} \leftarrow \mathbf{w} + \eta y_i x_i \quad b \leftarrow b + \eta y_i$$

with learning rate $\eta = 0.01$ for a maximum of 300 epochs.

Early stopping: Training halts as soon as a full epoch passes with zero misclassifications.

Convergence Theorem: If the data is linearly separable with margin γ and $\|x\| \leq R$, the perceptron makes at most:

$$M \leq \left(\frac{R}{\gamma}\right)^2 \quad \text{mistakes}$$

A larger margin γ (well-separated classes) guarantees fewer mistakes and faster convergence.

Decision Boundary: The learned boundary is the hyperplane $\mathbf{w}^\top x + b = 0$, i.e.

$$x_2 = -\frac{w_1 x_1 + b}{w_2}$$

3.2 Outputs and Results

- **Dataset A:** Convergence epoch reported (expected: few epochs). Test accuracy reported.
- **Dataset B:** Convergence epoch reported (may not converge). Test accuracy reported.

3.3 Plots and Explanations

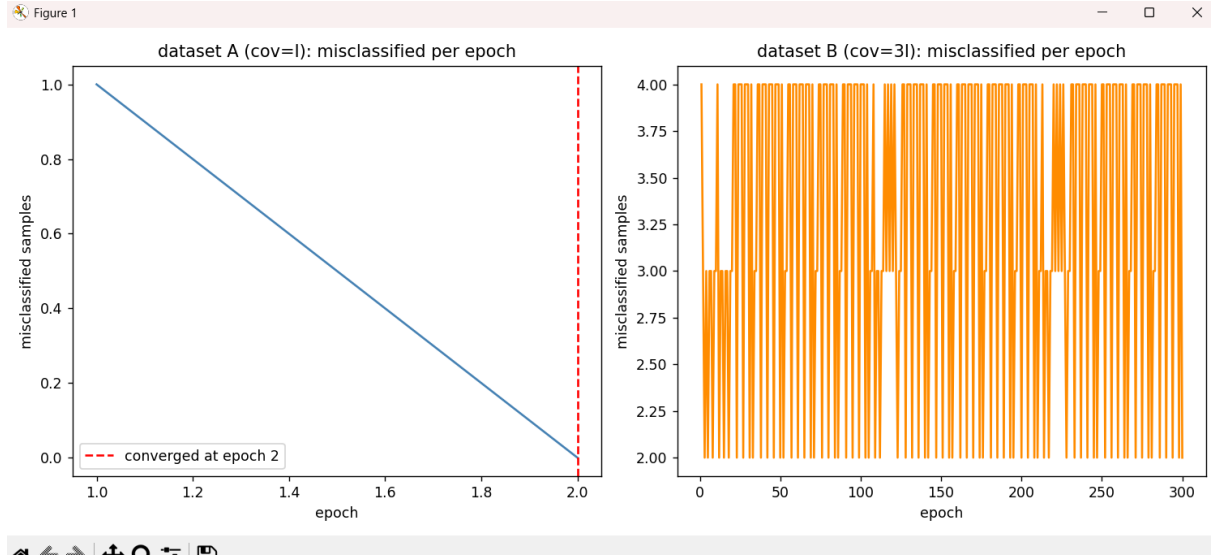


Figure 4: Misclassified Samples per Epoch: Dataset A (left) and Dataset B (right)

Observation (Dataset A): Misclassifications drop rapidly and reach zero within very few epochs. The classes are well-separated (large margin γ), so the convergence theorem bound $M \leq (R/\gamma)^2$ is small. The decision boundary settles cleanly between the two Gaussian clusters.

Observation (Dataset B): With covariance $3I$, the classes have much higher spread and overlap near the decision boundary. If the data is not perfectly linearly separable, the perceptron never converges and misclassifications oscillate throughout all 300 epochs. Even if technically separable, the reduced margin means significantly more mistakes before convergence. Test accuracy is lower due to the intrinsic class overlap.

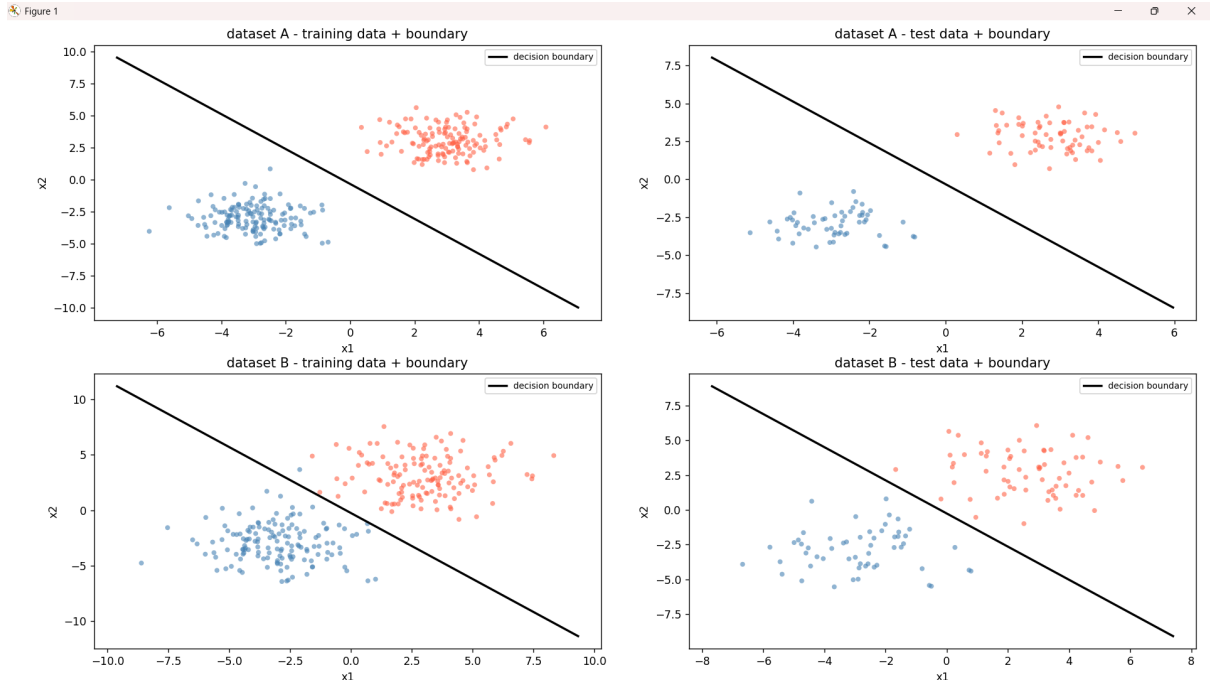


Figure 5: Final Decision Boundary on Training Data (left) and Test Data (right) for Datasets A and B

Observation: For Dataset A, the learned boundary is a clean diagonal line passing squarely between the two clusters on both train and test data. For Dataset B, the boundary still approximates the correct diagonal direction, but some points from both classes lie on the wrong side, reflecting the unavoidable overlap introduced by the larger covariance.

3.4 Trends and Justification

- **Dataset A ($\text{cov} = I$):** Data is linearly separable with a large margin. The perceptron converges in a small number of epochs. High test accuracy is expected because the decision boundary generalises well to unseen data from the same well-separated distributions.
- **Dataset B ($\text{cov} = 3I$):** Higher variance shrinks the effective margin γ between the classes. By the convergence bound, more mistakes are needed before any separating hyperplane is found. If the classes overlap, no perfect separator exists and the perceptron oscillates indefinitely. The test accuracy drops because some test points fall on the wrong side of any linear boundary due to the class overlap.
- **Key Insight:** The Rosenblatt Perceptron is guaranteed to converge *if and only if* the data is linearly separable. The speed of convergence is governed by the margin: convergence speed $\propto \gamma^2$. Larger covariance reduces γ , leading to slower or failed convergence and lower test accuracy.